

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Organic Chemistry Experimental Suite

Planning, Analysis and Evaluation · Multi-Step Aspirin Synthesis & Recrystallisation · Saponification Kinetics

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 — Planning, Analysis and Evaluation	DURATION:	1 Hour 15 Minutes	MAX MARKS:	30 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning the Preparative Synthesis, Purification, and Purity Analysis of Aspirin	9701/52/M/J/23/ Q1 · Syllabus 32.2 / 33.3 / 36.1	PLANNING	15
Question 2	Kinetics and Thermodynamic Analysis of the Alkaline Hydrolysis of Ethyl Benzoate	9701/51/O/N/23/Q 2 · Syllabus 26.1 / 33.2	ANALYSIS & EVALUATION	15

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QUESTION 1

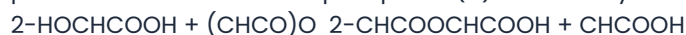
PLANNING

[Total: 15 Marks]

Planning the Preparative Synthesis, Purification, and Purity Analysis of Aspirin

Cambridge Syllabus Reference: 9701/52/M/J/23/Q1 · Syllabus 32.2 / 33.3 / 36.1

Aspirin (2-ethanoyloxybenzoic acid, $M_r = 180.2$) is prepared in the laboratory by the acylation of 2-hydroxybenzoic acid (salicylic acid, $M_r = 138.1$) using ethanoic anhydride ((CHCO)O, $M_r = 102.1$, density = 1.08 g cm^{-3}) in the presence of concentrated phosphoric(V) acid catalyst:



You are required to plan an experimental procedure to:

1. Synthesise crude aspirin starting from exactly 5.00 g of 2-hydroxybenzoic acid.
2. Purify the crude aspirin product by recrystallisation from a water/ethanol solvent mixture.
3. Calculate the maximum theoretical yield of aspirin.
4. Assess the purity of the recrystallised aspirin using capillary melting point determination and thin-layer chromatography (TLC).

You are provided with: solid 2-hydroxybenzoic acid, liquid ethanoic anhydride, concentrated HPO, ethanol, distilled water, 1% aqueous iron(III) chloride, TLC plates (silica gel on aluminium foil), UV lamp (254 nm), melting point apparatus, capillary tubes, Büchner funnel, vacuum filtration flask, and standard laboratory glassware.

- (a) Calculate the theoretical mass of aspirin produced from 5.00 g of 2-hydroxybenzoic acid, and calculate the minimum volume of ethanoic anhydride required if a 50% stoichiometric excess is used. [3]

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- (b) Describe the apparatus setup and procedure for heating the reaction mixture to ensure complete conversion without loss of volatile reagents. [3]

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- (c) Detail the recrystallisation procedure used to purify the crude aspirin, specifying solvent choice, filtration steps, and drying. [4]

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- (d) Describe how the melting point of the purified aspirin is determined, and state two characteristics of the melting point that confirm high purity. [3]

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- (e) Describe a chemical test using aqueous iron(III) chloride to confirm that the purified aspirin is free from unreacted 2-hydroxybenzoic acid. [2]

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QUESTION 2

ANALYSIS & EVALUATION

[Total: 15 Marks]

Kinetics and Thermodynamic Analysis of the Alkaline Hydrolysis of Ethyl Benzoate

Cambridge Syllabus Reference: 9701/51/O/N/23/Q2 · Syllabus 26.1 / 33.2

The alkaline hydrolysis (saponification) of ethyl benzoate is a second-order reaction:



Equal initial concentrations of ethyl benzoate and sodium hydroxide ($[\text{ester}] = [\text{OH}^-] = 0.0500 \text{ mol dm}^{-3}$) were mixed at 25.0°C .

At measured time intervals, 10.0 cm^3 samples were withdrawn using a pipette, immediately discharged into excess ice-cold water to quench the reaction, and titrated with standard $0.0200 \text{ mol dm}^{-3}$ hydrochloric acid to determine $[\text{OH}^-]$.

For equal initial reactant concentrations, the second-order integrated rate law is:

$$(1 / [\text{OH}^-]) = k \cdot t + (1 / [\text{OH}^-]_0)$$

where k is the rate constant in $\text{dm}^3 \text{ mol}^{-1} \text{ min}^{-1}$ and t is time in minutes.

- (a) The experimental data collected is displayed in the table below. Explain why the rate of reaction decreases as time progresses.

[2]

Sample	Time t / min	Titre of HCl V / cm^3	$[\text{OH}^-] / \text{mol dm}^{-3}$	$1 / [\text{OH}^-] / \text{dm}^3 \text{ mol}^{-1}$
1	0.0	25.00	0.0500	20.0
2	5.0	20.80	0.0416	24.0
3	10.0	17.85	0.0357	28.0
4	15.0	15.60	0.0312	32.1
5	20.0	13.90	0.0278	36.0
6	25.0	12.50	0.0250	40.0
7	30.0	11.35	0.0227	44.1
8	35.0	10.40	0.0208	48.1

- (b) A graph of $(1 / [\text{OH}^-])$ on the y-axis against time t / min on the x-axis was plotted.

[2]

- (i) State why this graph yields a straight line for a second-order reaction.
(ii) State what the y-intercept of the line represents.

- (c) Using the data from the table, calculate the gradient of the line of best fit and hence determine the rate constant k at 25°C , including its units.

[3]

- (d) Explain the physical principle of 'quenching' used in this experiment, and explain why discharging the aliquot into ice-cold water effectively freezes the reaction.

[4]

- (e) Suggest an alternative continuous instrumental technique to monitor the reaction rate without requiring sampling or chemical quenching, explaining your reasoning.

[4]

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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning the Preparative Synthesis, Purification, and Purity Analysis of Aspirin

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Moles of 2-hydroxybenzoic acid = $5.00 / 138.1 = 0.03621 \text{ mol}$ [1]; Theoretical mass of aspirin = $0.03621 \times 180.2 = 6.52 \text{ g}$ [1]; Moles of anhydride needed (50% excess) = $0.03621 \times 1.50 = 0.0543 \text{ mol}$; Mass = $0.0543 \times 102.1 = 5.54 \text{ g}$; Volume = $5.54 / 1.08 = 5.13 \text{ cm}^3$ (allow 5.1 to 5.2 cm^3) [1].	[3]
(b)	Add 5.00 g salicylic acid and 5.2 cm^3 ethanoic anhydride to a pear-shaped flask, add 5 drops of concentrated H_3PO_4 [1]; Fit a vertical Liebig condenser (reflux setup) with cold water entering at the bottom and leaving at the top [1]; Heat in a boiling water bath at $85\text{--}90^\circ\text{C}$ for 15 minutes to allow condensation of vapors back into the flask without loss of volatile reagents [1].	[3]
(c)	Dissolve crude product in the minimum volume of hot ethanol/water solvent mixture [1]; Filter while hot through fluted filter paper to remove any insoluble impurities [1]; Allow filtrate to cool slowly to room temperature, then chill in an ice bath to crystallise aspirin [1]; Filter crystals using a Büchner funnel under reduced pressure (vacuum filtration), wash with small volume of ice-cold solvent, and dry between filter papers / in a desiccator [1].	[4]
(d)	Pack a small amount of dry powder into a thin capillary tube sealed at one end [1]; Place in melting point apparatus (e.g. electrical block or Thiele tube), heat slowly near expected m.p. ($138\text{--}140^\circ\text{C}$), and record initial melting and complete melting temperatures [1]; High purity is confirmed by: (1) sharp melting point over a narrow range ($< 1\text{--}2^\circ\text{C}$), and (2) melting point value matches the literature value ($138\text{--}140^\circ\text{C}$) without depression [1].	[3]
(e)	Add a few drops of neutral $\text{FeCl}_3(\text{aq})$ to an aqueous solution of the purified aspirin [1]; Unreacted 2-hydroxybenzoic acid contains a phenolic --OH group and gives an intense purple/violet coloration; absence of purple color (solution remains yellow) confirms that no salicylic acid impurity is present [1].	[2]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Kinetics and Thermodynamic Analysis of the Alkaline Hydrolysis of Ethyl Benzoate

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	As the reaction proceeds, reactant concentrations ($[\text{ester}]$ and $[\text{OH}^-]$) decrease steadily [1]; The collision frequency between ester molecules and hydroxide ions decreases, causing rate ($\text{rate} = k[\text{ester}][\text{OH}^-]$) to drop [1].	[2]
(b)	(i) The equation $1/[\text{OH}^-] = kt + 1/[\text{OH}^-]_0$ is of the linear form $y = mx + c$, where slope $m = k$ and y-intercept $c = 1/[\text{OH}^-]_0$ [1]; (ii) The y-intercept represents $1/[\text{OH}^-]_0$ (the reciprocal of the initial hydroxide ion concentration at $t = 0$) [1].	[2]
(c)	Two points on line selected (e.g. (0.0, 20.0) and (35.0, 48.1)) [1]; Gradient = $y/x = (48.1 - 20.0) / (35.0 - 0.0) = 28.1 / 35.0 = 0.803 \text{ dm}^3 \text{ mol}^{-1} \text{ min}^{-1}$ [1]; Rate constant $k = 0.803 \text{ dm}^3 \text{ mol}^{-1} \text{ min}^{-1}$ (or $0.0134 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$) with correct units [1].	[3]
(d)	Quenching rapidly stops or dramatically slows down the reaction so that the concentration of reactants does not change during the subsequent titration [1]; Dilution with a large volume of water drastically lowers the concentrations of ester and OH^- , reducing collision frequency [1]; Low temperature (near 0°C) significantly reduces average kinetic energy, drastically lowering the fraction of molecules with energy $E \geq E_a$ [2].	[4]
(e)	Electrical conductivity measurement [1]; High-mobility hydroxide ions (OH^-) are replaced one-for-one by bulky benzoate ions ($\text{C}_6\text{H}_5\text{COO}^-$) of much lower ionic mobility [1]; The electrical conductivity of the solution decreases steadily and continuously over time, allowing real-time monitoring without disturbing the equilibrium [2].	[4]